



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

2002
YEAR 11 PRELIMINARY
EXAMINATION

Mathematics

General Instructions

- Reading time — 5 minutes
- Working time — 90 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question

Total marks — 80

- Attempt questions 1–5
- All questions are of equal value

Examiner: B Opferkuch

QUESTION 1.**(16 marks)****Start a new booklet****Marks**

- (a) Calculate $\sqrt{\frac{19}{4\pi}}$ correct to three decimal places. **1**
- (b) Simplify $\sqrt{48} + \sqrt{108}$. **2**
- (c) Factorise $4x^2 - 1$. **1**
- (d) Express $(27x^6)^{\frac{1}{3}}x^{-3}$ in the simplest form, without the use of negative indices. **2**
- (e) Express the recurring decimal $0.\dot{6}\dot{7}$ as a fraction in its simplest form. **1**
- (f) Write down, in surd form, the value of $\cos 210^\circ$. **1**
- (g) Find the value of x , if $\log_x 125 = 3$. **2**
- (h) Solve $|5x - 4| = 9$ **2**
- (i) If $g(x) = \begin{cases} 2 - x^3 & \text{if } x > 0 \\ x^2 + 1 & \text{if } x \leq 0 \end{cases}$ **2**
- Find the value of $g(3) + g(-4) - g(0)$.
- (j) Graph on a number line the solution of the inequality $-2 < 4 - 2x \leq 5$ **2**
- (k) Solve $2^{5-3x} = 1$ **2**

QUESTION 2. (16 marks) **Start a new booklet**

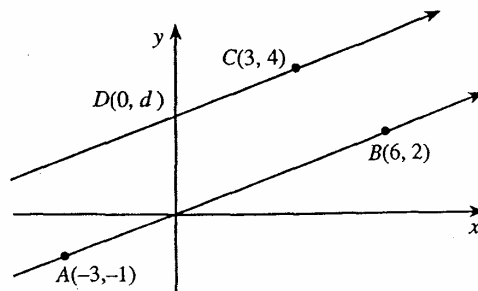
Marks

- (a) Factorise
- (i) $x^2 - x - 6$ 1
 - (ii) $3x^2 + 4x - 7$ 2
 - (iii) $8a^3 - 27$ 2
- (b) Consider the function $f(x) = \sqrt{4 - x^2}$.
- (i) Show that the function is even.
 - (ii) What is the range of $f(x)$?
 - (iii) Sketch the curve.
- (c) If α and β are the roots of $2x^2 - 5x + 1 = 0$, find the values of
- (i) $\alpha + \beta$
 - (ii) $\alpha\beta$
 - (iii) $\frac{2}{\alpha} + \frac{2}{\beta}$
- (d) Simplify $\cot A \sec A \sin A$ 2

QUESTION 3. (16 marks) **Start a new booklet**

Marks

(a)



The diagram above shows the points $A(-3, -1)$, $B(6, 2)$, $C(3, 4)$ and $D(0, d)$.

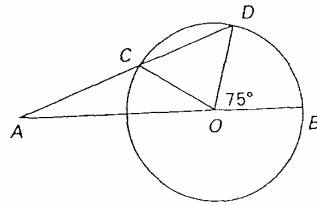
Copy the diagram to your answer booklet.
On your diagram indicate the given information.

- | | | |
|-------|--|---|
| (i) | Find the gradient of AB . | 1 |
| (ii) | Find the value of d if CD and AB are parallel. | 1 |
| (iii) | Show that the equation of AB is $x - 3y = 0$. | 2 |
| (iv) | Find the perpendicular distance between C and the line AB . | 2 |
| (v) | Prove that $AB = 3CD$. | 3 |
| (vi) | Find the area of the quadrilateral $ABCD$. | 1 |
| | | |
| (b) | In triangle ABC , $AB = 5\text{cm}$, $BC = 3\text{cm}$, and $\angle CAB = 35^\circ 11'$.
Find the possible values for $\angle ACB$. | 3 |

QUESTION 3. continued

Marks

(c)



In this figure, AOB and ACD are straight lines.

O is the centre of circle radius r . $AC = DO$ and $\angle DOB = 75^\circ$.

If $\angle CAO = x$:

- (i) Show that $\angle DCO = 2x$.
- (ii) Show that $\angle DOB = 3x = 75^\circ$

1

2

QUESTION 4. (16 marks) **Start a new booklet**

Marks

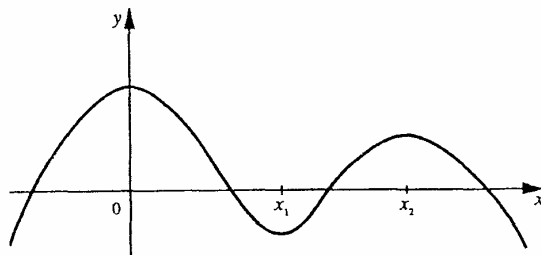
(a) Differentiate the following with respect to x :

- | | | |
|-------|-----------------|---|
| (i) | $3x^4 - 4x + 7$ | 1 |
| (ii) | $\frac{2}{x^3}$ | 1 |
| (iii) | $x\sqrt{x}$ | 1 |

(b) (i) Use the product rule to differentiate $y = 5x^3(4x - 3)^5$. 2

(ii) Differentiate $y = \frac{x^2 - 2}{x + 1}$ by using the quotient rule. 3

(c) Sketch the gradient function of the following curve. 2



(d) Find the equation of the normal to the parabola $y = \frac{1}{5}x^2$ at the point $(2, \frac{4}{5})$. 3

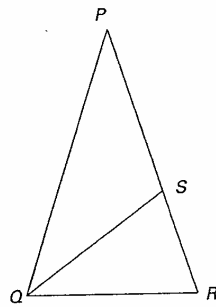
(e) (i) Write down the discriminant of $3x^2 + 2x + k$. 1

(ii) For what values of k does $3x^2 + 2x + k$ have real roots? 2

QUESTION 5. (16 marks) **Start a new booklet**

Marks

- (a) The focus of a parabola is $S(0,4)$ and its directrix is the line $y = -2$. 4
- (i) Sketch the parabola and indicate the coordinates of the vertex V .
- (ii) Write down the focal length of the parabola.
- (iii) Find the equation of the parabola.
- (b) Consider $\log_a b^2 + \log_a b^3 - \log_a b^4 = 1$. 2
Find the relationship between a and b .
- (c) In $\triangle PQR$, $\angle QPR = x$, $\angle PQR = 2x$ and QS bisects $\angle PQR$, meeting PR at S .



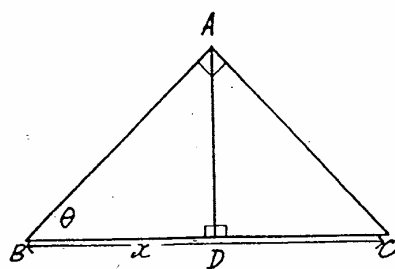
- (i) Prove that $\triangle QSR \sim \triangle PQR$. 3
- (ii) If $QR = 6\text{cm}$ and $PQ = 10\frac{2}{3}\text{cm}$, find PR . 3

QUESTION 5. continued

Marks

- (d) The diagram shows a right angled triangle ABC . $\angle ABC = \theta$.
The line AD is perpendicular to BC and $BC = x$ units.

4



- (i) show that $BA = x \cos \theta$
- (ii) show that $BD = x \cos^2 \theta$
- (iii) and hence that $DC = x \sin^2 \theta$

End of paper



SYDNEY BOYS HIGH SCHOOL
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SEPTEMBER 2002

YEARLY EXAMINATION

YEAR 11

Mathematics

Sample Solutions

Question 1

$$(a) \quad \sqrt{\frac{19}{4\pi}} \approx 1.22962269\dots$$

$$= 1.230 \quad (3 \text{ d.p.})$$

$$(b) \quad \sqrt{48} + \sqrt{108} = 4\sqrt{3} + 6\sqrt{3} = 10\sqrt{3}$$

$$(c) \quad 4x^2 - 1 = (2x - 1)(2x + 1)$$

$$(d) \quad (27x^6)^{\frac{1}{3}} x^{-3} = 3 \times x^2 \times x^{-3} = 3x^{-1} = \frac{3}{x}$$

$$(e) \quad 0.\dot{6}\dot{7} = \frac{67}{99}$$

$$(f) \quad \cos 210^\circ = \cos(180 + 30)^\circ = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$$

$$(g) \quad \log_x 125 = 3 \Leftrightarrow x^3 = 125$$

$$\therefore x = \sqrt[3]{125} = 5$$

$$(h) \quad |5x - 4| = 9$$

$$\therefore 5x - 4 = 9, 5x - 4 = -9$$

$$\therefore x = 2.6, -1$$

$$(i) \quad g(x) = \begin{cases} 2 - x^3 & \text{if } x > 0 \\ x^2 + 1 & \text{if } x \leq 0 \end{cases}$$

$$\therefore g(3) + g(-4) - g(0) = (2 - 3^3) + [(-4)^2 + 1] - [0^2 + 1] = -9$$

$$(j) \quad -2 < 4 - 2x \leq 5 \Rightarrow -6 < -2x \leq 1$$

$$\therefore 3 > x \geq -\frac{1}{2}$$



$$(k) \quad 2^{5-3x} = 1 = 2^0$$

$$\therefore 5 - 3x = 0$$

$$\therefore x = \frac{5}{3}$$

Question 2

(a) (i) $x^2 - x - 6 = (x - 3)(x + 2)$

(ii) $3x^2 + 4x - 7$

$(3 \times -7 = -21 \Rightarrow a + b = 4, ab = -21)$

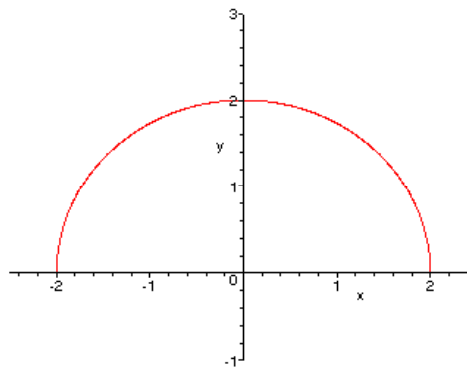
$3x^2 + 4x - 7 = 3x^2 + 7x - 3x - 7 = x(3x + 7) - (3x + 7) = (x - 1)(3x + 7)$

(iii) $8a^3 - 27 = (2a)^3 - 3^3 = (2a - 3)[(2a)^2 + (2a) \times 3 + 3^2] = (2a - 3)(4a^2 + 6a + 9)$

(b) (i) $f(-x) = \sqrt{4 - (-x)^2} = \sqrt{4 - x^2} = f(x)$

(ii) $0 \leq y \leq 2$

(iii)



(c) $2x^2 - 5x + 1 = 0$

(i) $\alpha + \beta = \frac{5}{2}$

(ii) $\alpha\beta = \frac{1}{2}$

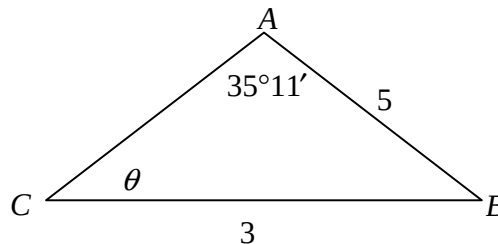
(iii) $\frac{2}{\alpha} + \frac{2}{\beta} = \frac{2(\alpha + \beta)}{\alpha\beta} = \frac{2 \times \frac{5}{2}}{\frac{1}{2}} = 10$

(d) $\cot A \times \sec A \times \sin A = \frac{\cos A}{\sin A} \times \frac{1}{\cos A} \times \sin A = 1$

Question 3

- (a) (i) $m_{AB} = \frac{-1-2}{-3-6} = \frac{1}{3}$
- (ii) $AB \parallel CD \Rightarrow m_{AB} = m_{CD}$
 $m_{CD} = \frac{d-4}{0-3} = -\frac{d-4}{3}$
 $\therefore -\frac{d-4}{3} = \frac{1}{3} \Rightarrow -d+4=1$
 $\therefore d=3$
- (iii) $y-y_1 = m_{AB}(x-x_1)$
 $\therefore y-2 = \frac{1}{3}(x-6) \Rightarrow 3y-6 = x-6$
 $\therefore x-3y=0$
- (iv) $d_{\perp} = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$
 $= \frac{|1 \times 3 + (-3) \times 4 + 0|}{\sqrt{1^2 + (-3)^2}} = \frac{9}{\sqrt{10}}$
- (v) $AB = \sqrt{(-3-6)^2 + (-1-2)^2} = \sqrt{81+9} = \sqrt{90} = 3\sqrt{10}$
 $CD = \sqrt{(0-3)^2 + (3-4)^2} = \sqrt{9+1} = \sqrt{10}$
 $\therefore AB = 3CD$
- (vi) $ABCD$ is a trapezium
 $\text{Area} = \frac{1}{2} \times d_{\perp} \times (CD + AB) = \frac{1}{2} \times \frac{9}{\sqrt{10}} \times 4\sqrt{10} = 18 \text{ units squared}$

(b)



$$\frac{\sin \theta}{5} = \frac{\sin 35^{\circ}11'}{3} \Rightarrow \sin \theta = \frac{5 \sin 35^{\circ}11'}{3} \approx 0.960324...$$

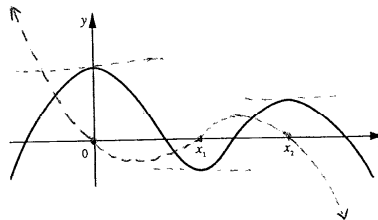
$$\therefore \theta = 73^{\circ}48', 106^{\circ}12'$$

- (c) (i) $OD = OC \Rightarrow AC = OC \therefore \angle DCO = 2x$ (Exterior angle theorem)
- (ii) $\angle CDO = 2x \Rightarrow \angle COD = 180 - 4x$
 $\therefore \angle DOB = 180 - (180 - 4x + x) = 3x$ (Straight angle)

Question 4

- (a) (i) $\frac{d}{dx}(3x^4 - 4x + 7) = 12x^3 - 4$
- (ii) $\frac{d}{dx}\left(\frac{2}{x^3}\right) = \frac{d}{dx}(2x^{-3}) = -6x^{-4} = -\frac{6}{x^4}$
- (iii) $\frac{d}{dx}(x\sqrt{x}) = \frac{d}{dx}\left(x^{\frac{3}{2}}\right) = \frac{3}{2}x^{\frac{1}{2}} = \frac{3\sqrt{x}}{2}$
- (b) (i) $\frac{dy}{dx} = (4x-3)^5 \times \frac{d}{dx}(5x^3) + 5x^3 \times \frac{d}{dx}((4x-3)^5)$
 $= 15x^2(4x-3)^5 + 5x^3 \times 5(4x-3)^4 \times 4$
 $= 5x^2(4x-3)^4 [3(4x-3) + 20x] = 5x^2(32x-9)(4x-3)^4$
- (ii) $\frac{dy}{dx} = \frac{(x+1) \times \frac{d}{dx}(x^2-2) - (x^2-2) \times \frac{d}{dx}(x+1)}{(x+1)^2}$
 $= \frac{(x+1) \times 2x - (x^2-2)}{(x+1)^2} = \frac{x^2+2x+2}{(x+1)^2}$

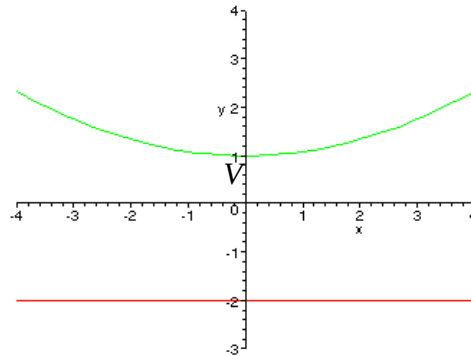
- (c) At $x=0, x_1, x_2$ the tangents are horizontal so $y' = 0$
 For $x < 0$, $x_1 < x < x_2$ y is increasing so $y' > 0$
 For $0 < x < x_1, x > x_2$ y is decreasing so $y' < 0$
 A possible graph could be



- (d) $y = \frac{x^2}{5} \Rightarrow \frac{dy}{dx} = \frac{2x}{5} \quad \therefore \frac{dy}{dx}_{x=2} = \frac{4}{5}$
 So the tangent has a gradient of $4/5$, so the normal has gradient $-5/4$
 So the normal has equation $y - \frac{4}{5} = -\frac{5}{4}(x-2)$
 $20y - 16 = -25(x-2) \quad \therefore 25x + 20y - 66 = 0$
- (e) (i) $\Delta = (2)^2 - 4 \times 3 \times k = 4 - 12k = 4(1-3k)$
- (ii) Real roots if $\Delta \geq 0 \Rightarrow 4(1-3k) \geq 0 \quad \therefore k \leq 1/3$

Question 5

- (a) (i) Vertex is midway between focus and directrix
 $V(0,1)$ and so $a = 3$



- (ii) Focal length $= |a| = 3$
- (iii) $(x-h)^2 = 4a(y-k) \Rightarrow x^2 = 4 \times 3 \times (y-1)$
 $x^2 = 12(y-1)$
- (b) $\log_a b^2 + \log_a b^3 - \log_a b^4 = 1$
 $\therefore \log_a \left(\frac{b^2 \times b^3}{b^4} \right) = 1 \Rightarrow \log_a b = 1$
 $\therefore a^1 = b$
 $\therefore a = b$
- (c) (i) $\because QS$ bisects $\angle PQR \Rightarrow \angle RQS = \angle SQP = x$
 $\angle QSR = \angle SQP + \angle SPQ = x + x = 2x$
 So $\triangle QSR \parallel \triangle PQR$ (equiangular)
 $[\angle PQR = \angle QSR = 2x, \angle QPR = \angle SQR = x]$
- (ii) Let $a = PR, b = SR, c = QS$
 $\frac{PR}{QR} = \frac{QR}{SR} \Rightarrow \frac{a}{6} = \frac{6}{b} \Rightarrow b = \frac{36}{a}$
 $\frac{PR}{QR} = \frac{PQ}{QS} \Rightarrow \frac{a}{6} = \frac{10\frac{2}{3}}{c} \Rightarrow c = \frac{64}{a}$
 Now $QS = SP$ ($\triangle QSP$ is isosceles) and $PR = SP + SR$
 So $PR = a = b + c$
 $\therefore a = \frac{36}{a} + \frac{64}{a} \Rightarrow a^2 = 100$
 $\therefore a = PR = 10$
- (d) (i) In $\triangle BAC$ $\cos \theta = \frac{AB}{x} \Rightarrow AB = x \cos \theta$
- (ii) In $\triangle ABD$ $\cos \theta = \frac{BD}{AB} \Rightarrow DB = AB \cos \theta = (x \cos \theta) \cos \theta = x \cos^2 \theta$
- (iii) $DC = BC - BD = x - x \cos^2 \theta = x(1 - \cos^2 \theta) = x \sin^2 \theta$
 $\therefore \sin^2 \theta + \cos^2 \theta = 1$